

**Title:**

**The health and economic value of Artificial Intelligence in a resource-constrained diagnostic pathway of prostate cancer in the Netherlands – a discrete event simulation model**

**Authors:**

Pim W.M. van Dorst<sup>1\*</sup> MSc, Stefan J. Fransen<sup>2\*</sup> MSc, Thijs Vlutters<sup>3</sup> MSc, Marco H. Blanker<sup>4</sup> MD, PhD, Pim J. van Leeuwen<sup>5</sup> MD, PhD, Shafak Al-Uwini<sup>6</sup> MD, PhD, Igle-Jan de Jong<sup>2</sup> MD, PhD, Hendrik Koffijberg<sup>7</sup> PhD, Thomas C. Kwee<sup>2</sup> MD, PhD, Antoinette D.I. van Asselt <sup>\*\*1,8</sup> PhD, Derya Yakar <sup>\*\*2</sup> MD PhD

\* These authors contributed equally and share the first authorship

\*\* These authors contributed equally and share the last authorship

**Affiliations:**

1. University of Groningen, University Medical Center Groningen, Department of Health Sciences
2. University of Groningen, University Medical Center Groningen, Department of Radiology
3. University of Groningen, Faculty of Economics and Business
4. University of Groningen, University Medical Center Groningen, department of Primary and Long-term Care
5. Antoni van Leeuwenhoek – Netherlands Cancer Institute, Amsterdam, The Netherlands, Department of Urology.
6. University of Groningen, University Medical Center Groningen, Department of Radiation Oncology
7. Section Health Technology and Services Research, TechMed Centre, Faculty of Behavioural, Management, and Social Sciences, University of Twente, Enschede, The Netherlands
8. University of Groningen, University Medical Center Groningen, Department of Epidemiology

**Journal:**

PharmacoEconomics

**Corresponding author:**

Pim W.M. van Dorst MSc, Unit of Global Health, Department of Health Sciences, University Medical Center Groningen, PO Box 30.001, 9700 RB Groningen, The Netherlands.

e-mail: [w.m.van.dorst@umcg.nl](mailto:w.m.van.dorst@umcg.nl)

## Appendix 1 - CHEERS 2022 Checklist

| Topic                                | No. | Item                                                                                                                            | Location where item is reported |
|--------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Title</b>                         |     |                                                                                                                                 |                                 |
|                                      | 1   | Identify the study as an economic evaluation and specify the interventions being compared.                                      | Page 1                          |
| <b>Abstract</b>                      |     |                                                                                                                                 |                                 |
|                                      | 2   | Provide a structured summary that highlights context, key methods, results, and alternative analyses.                           | Page 2                          |
| <b>Introduction</b>                  |     |                                                                                                                                 |                                 |
| <b>Background and objectives</b>     | 3   | Give the context for the study, the study question, and its practical relevance for decision making in policy or practice.      | Page 3                          |
| <b>Methods</b>                       |     |                                                                                                                                 |                                 |
| <b>Health economic analysis plan</b> | 4   | Indicate whether a health economic analysis plan was developed and where available.                                             | NA                              |
| <b>Study population</b>              | 5   | Describe characteristics of the study population (such as age range, demographics, socioeconomic, or clinical characteristics). | Page 4                          |
| <b>Setting and location</b>          | 6   | Provide relevant contextual information that may influence findings.                                                            | Page 4                          |
| <b>Comparators</b>                   | 7   | Describe the interventions or strategies being compared and why chosen.                                                         | Page 7-8                        |
| <b>Perspective</b>                   | 8   | State the perspective(s) adopted by the study and why chosen.                                                                   | Page 8                          |
| <b>Time horizon</b>                  | 9   | State the time horizon for the study and why appropriate.                                                                       | Page 4                          |
| <b>Discount rate</b>                 | 10  | Report the discount rate(s) and reason chosen.                                                                                  | Page 8 and page 11              |
| <b>Selection of outcomes</b>         | 11  | Describe what outcomes were used as the measure(s) of benefit(s) and harm(s).                                                   | Page 12-13                      |
| <b>Measurement of outcomes</b>       | 12  | Describe how outcomes used to capture benefit(s) and harm(s) were measured.                                                     | Page 12-13                      |
| <b>Valuation of outcomes</b>         | 13  | Describe the population and methods used to measure and value outcomes.                                                         | Page 12-13                      |

| Topic                                                                        | No. | Item                                                                                                                                                                          | Location where item is reported |
|------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Measurement and valuation of resources and costs</b>                      | 14  | Describe how costs were valued.                                                                                                                                               | Page 8-12                       |
| <b>Currency, price date, and conversion</b>                                  | 15  | Report the dates of the estimated resource quantities and unit costs, plus the currency and year of conversion.                                                               | Page 8                          |
| <b>Rationale and description of model</b>                                    | 16  | If modelling is used, describe in detail and why used. Report if the model is publicly available and where it can be accessed.                                                | Page 4-7                        |
| <b>Analytics and assumptions</b>                                             | 17  | Describe any methods for analysing or statistically transforming data, any extrapolation methods, and approaches for validating any model used.                               | Page 4-12                       |
| <b>Characterising heterogeneity</b>                                          | 18  | Describe any methods used for estimating how the results of the study vary for subgroups.                                                                                     | NA                              |
| <b>Characterising distributional effects</b>                                 | 19  | Describe how impacts are distributed across different individuals or adjustments made to reflect priority populations.                                                        | NA                              |
| <b>Characterising uncertainty</b>                                            | 20  | Describe methods to characterise any sources of uncertainty in the analysis.                                                                                                  | Page 13                         |
| <b>Approach to engagement with patients and others affected by the study</b> | 21  | Describe any approaches to engage patients or service recipients, the general public, communities, or stakeholders (such as clinicians or payers) in the design of the study. | NA                              |
| <b>Results</b>                                                               |     |                                                                                                                                                                               |                                 |
| <b>Study parameters</b>                                                      | 22  | Report all analytic inputs (such as values, ranges, references) including uncertainty or distributional assumptions.                                                          | Page 4-12                       |
| <b>Summary of main results</b>                                               | 23  | Report the mean values for the main categories of costs and outcomes of interest and summarise them in the most appropriate overall measure.                                  | Page 13-17                      |
| <b>Effect of uncertainty</b>                                                 | 24  | Describe how uncertainty about analytic judgments, inputs, or projections affect findings. Report the effect of choice of discount rate and time horizon, if applicable.      | Page 16-17                      |
| <b>Effect of engagement with patients and others affected by the study</b>   | 25  | Report on any difference patient/service recipient, general public, community, or stakeholder involvement made to the approach or findings of the study                       | NA                              |
| <b>Discussion</b>                                                            |     |                                                                                                                                                                               |                                 |

| Topic                                                                       | No. | Item                                                                                                                                       | Location where item is reported |
|-----------------------------------------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Study findings, limitations, generalisability, and current knowledge</b> | 26  | Report key findings, limitations, ethical or equity considerations not captured, and how these could affect patients, policy, or practice. | Page 17-19                      |
| <b>Other relevant information</b>                                           |     |                                                                                                                                            |                                 |
| <b>Source of funding</b>                                                    | 27  | Describe how the study was funded and any role of the funder in the identification, design, conduct, and reporting of the analysis         | NA                              |
| <b>Conflicts of interest</b>                                                | 28  | Report authors conflicts of interest according to journal or International Committee of Medical Journal Editors requirements.              | Page 21                         |

*From:* Husereau D, Drummond M, Augustovski F, et al. Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) Explanation and Elaboration: A Report of the ISPOR CHEERS II Good Practices Task Force. Value Health 2022;25.  
[doi:10.1016/j.jval.2021.10.008](https://doi.org/10.1016/j.jval.2021.10.008)